



IPB University
— Bogor Indonesia —

Computer Science Study Program
School of Data Science, Mathematics and Informatics



Kecerdasan Buatan **Kuliah 6 – Logical Agents dan Propositional Logic**

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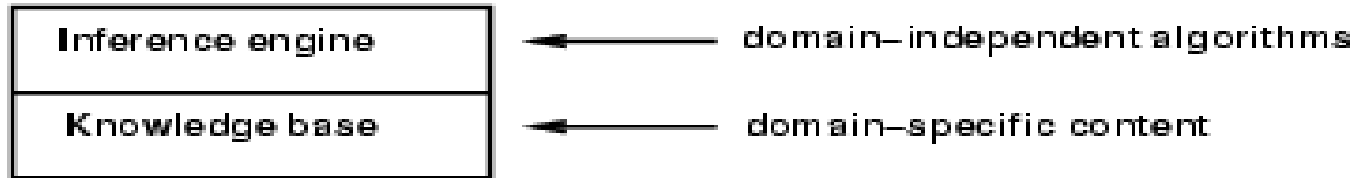
Outline

- Knowledge-based agents
- Wumpus world
- Logic in general - models and entailment
- Propositional (Boolean) logic
- Equivalence, validity, satisfiability

Knowledge bases



- Knowledge base = set of **sentences** in a **formal** language
- **Declarative** approach to building an agent (or other system):
 - Tell it what it needs to know
- Then it can **Ask** itself what to do - answers should follow from the KB
- Agents can be viewed at the **knowledge level**
i.e., what they know, regardless of how implemented
- Or at the **implementation level**
 - i.e., data structures in KB and algorithms that manipulate them



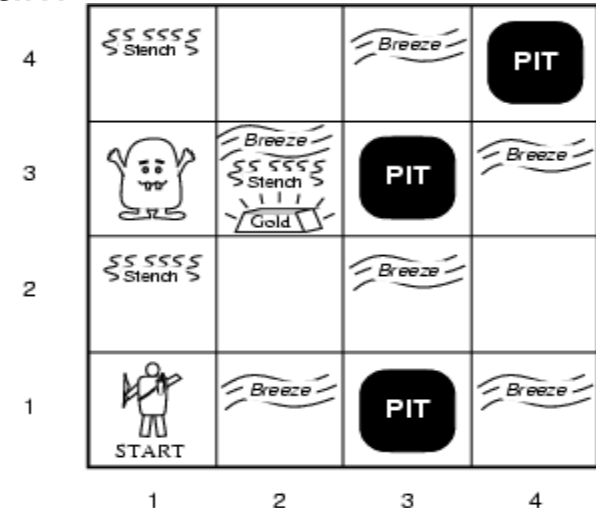
```
function KB-AGENT(percept) returns an action  
  static: KB, a knowledge base  
           t, a counter, initially 0, indicating time  
  
  TELL(KB, MAKE-PERCEPT-SENTENCE(percept, t))  
  action  $\leftarrow$  ASK(KB, MAKE-ACTION-QUERY(t))  
  TELL(KB, MAKE-ACTION-SENTENCE(action, t))  
  t  $\leftarrow$  t + 1  
  return action
```

- The agent must be able to:
 - Represent states, actions, etc
 - Incorporate new percepts
 - Update internal representations of the world
 - Deduce hidden properties of the world
 - Deduce appropriate actions

Wumpus World



- Performance measure
 - gold +1000, death -1000
 - -1 per step, -10 for using the arrow
- **Actuators:** Left turn, Right turn, Forward, Grab, Release, Shoot
- **Sensors:** Stench, Breeze, Glitter, Bump, Scream
- **Environment**
 - Squares adjacent to wumpus are smelly
 - Squares adjacent to pit are breezy
 - Glitter iff gold is in the same square
 - Shooting kills wumpus if you are facing it
 - Shooting uses up the only arrow
 - Grabbing picks up gold if in same square
 - Releasing drops the gold in same square



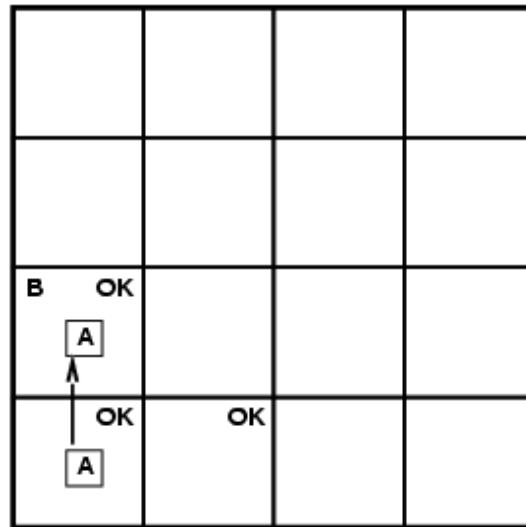
Wumpus world characterization

- Fully Observable No – only **local** perception
- Deterministic Yes – outcomes exactly specified
- Episodic No – sequential at the level of actions
- Static Yes – Wumpus and Pits do not move
- Discrete Yes
- Single-agent? Yes – Wumpus is essentially a natural feature

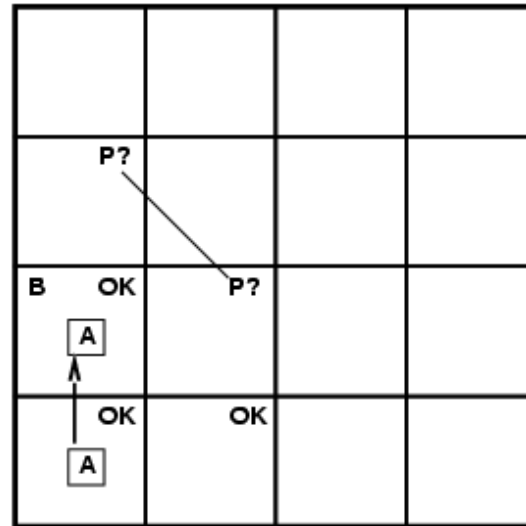
Exploring a Wumpus world

OK			
OK A	OK		

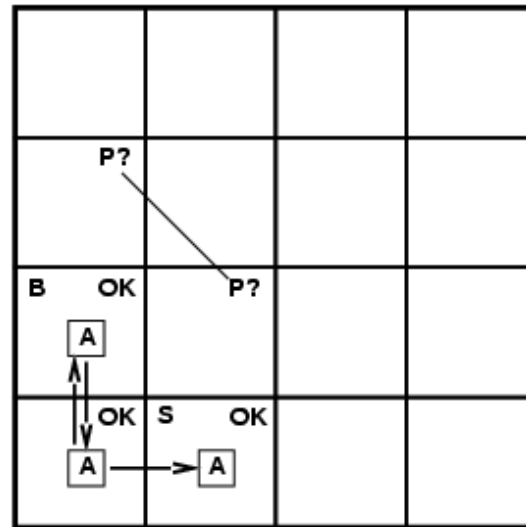
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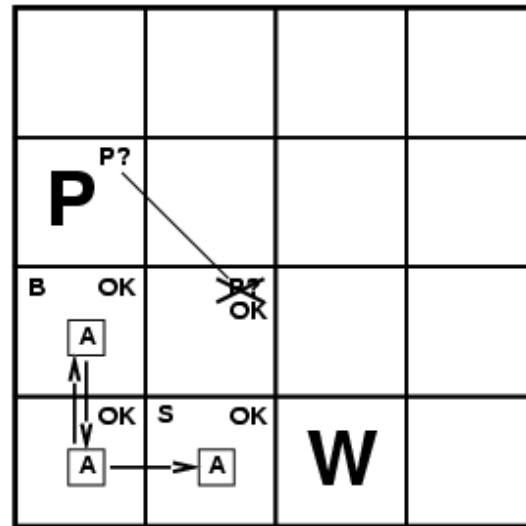
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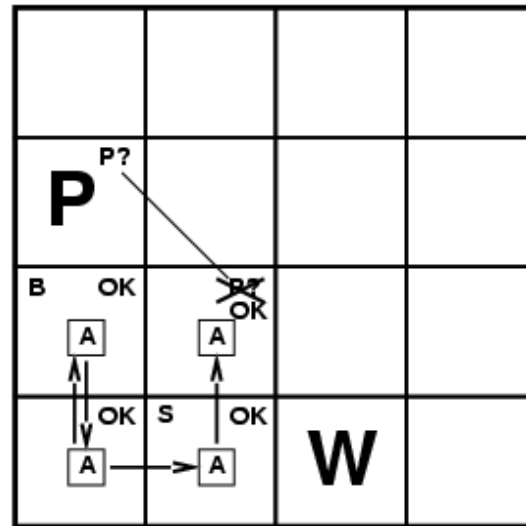
Exploring a Wumpus world



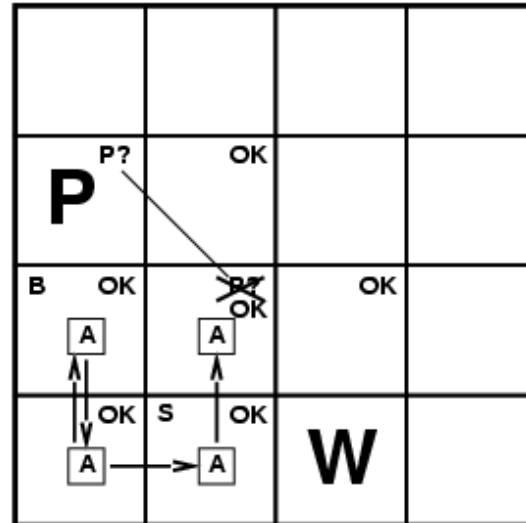
Exploring a Wumpus world



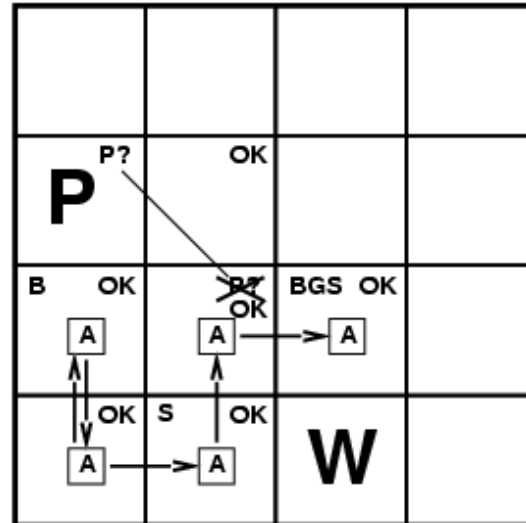
Exploring a Wumpus world



Exploring a Wumpus world



Exploring a Wumpus world



Logic in general

- **Logics** are formal languages for representing information such that conclusions can be drawn
- **Syntax** defines the sentences in the language
- **Semantics** define the "meaning" of sentences;
 - i.e., define **truth** of a sentence in a world
- E.g., the language of arithmetic
 - $x+2 \geq y$ is a sentence; $x^2+y > \{\}$ is not a sentence
 - $x+2 \geq y$ is true iff the number $x+2$ is no less than the number y
 - $x+2 \geq y$ is true in a world where $x = 7, y = 1$
 - $x+2 \geq y$ is false in a world where $x = 0, y = 6$

Entailment

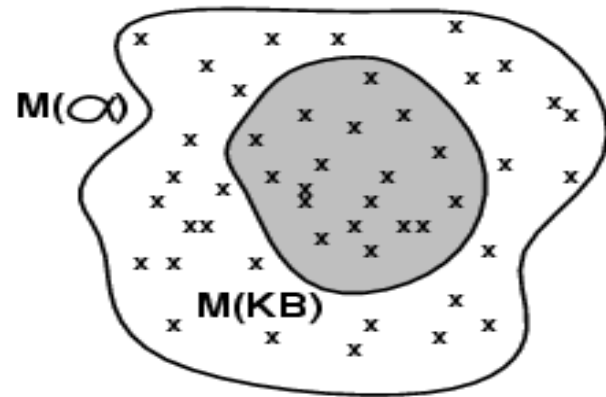
- Entailment means that one thing **follows from** another:

$$KB \models \alpha$$

- Knowledge base KB entails sentence α if and only if α is true in all worlds where KB is true
 - E.g., the KB containing “the Giants won” and “the Reds won” entails “Either the Giants won or the Reds won”
- Entailment is a relationship between sentences (i.e., syntax) that is based on semantics

Models

- Logicians typically think in terms of **models**, which are formally structured worlds with respect to which truth can be evaluated
- We say m is a **model of** a sentence α if α is true in m
- $M(\alpha)$ is the set of all models of α
- Then $KB \models \alpha$ iff $M(KB) \subseteq M(\alpha)$
 - E.g. $KB = \text{Giants won and Reds won}$
 $\alpha = \text{Giants won}$

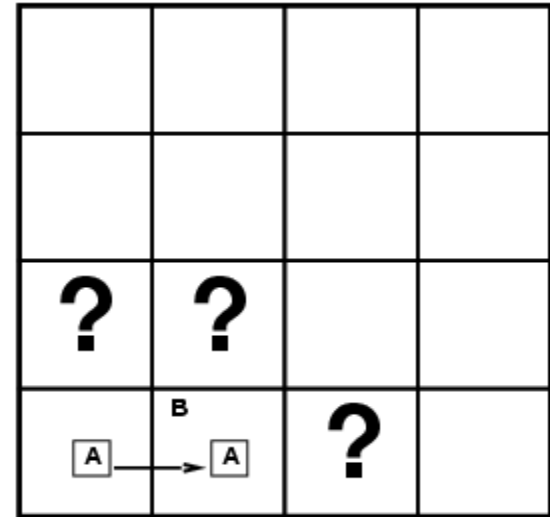


Entailment in the Wumpus world

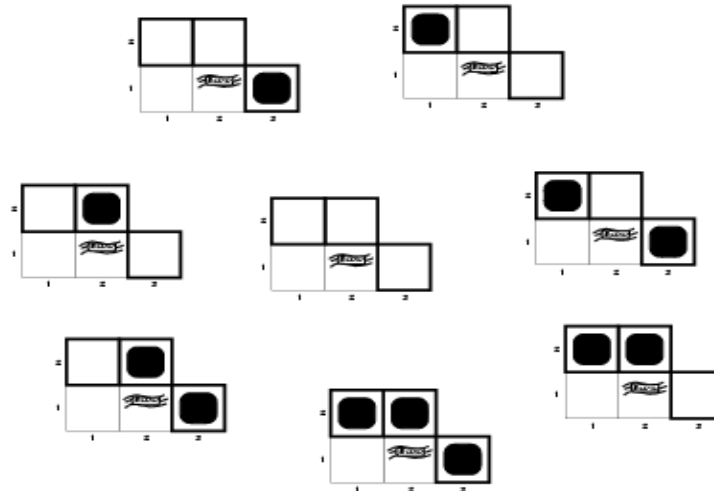
Situation after detecting nothing in $[1,1]$,
moving right, breeze in $[2,1]$

Consider possible models for KB
assuming only pits

3 Boolean choices $\Rightarrow$ 8 possible models

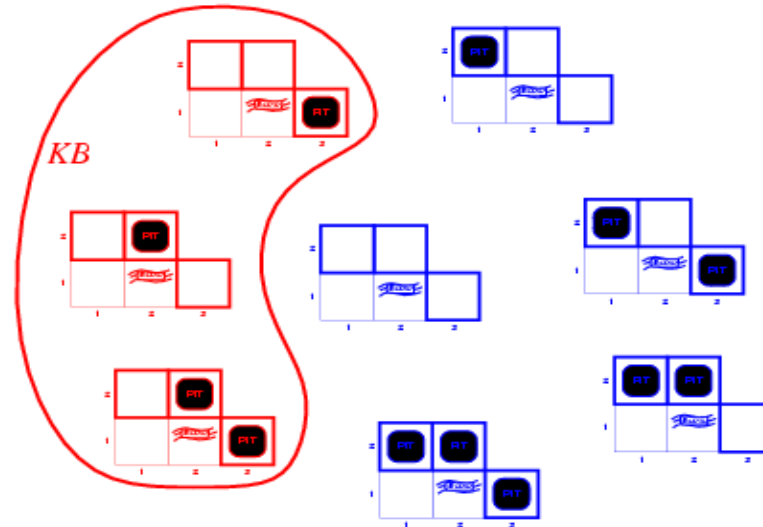


Wumpus models



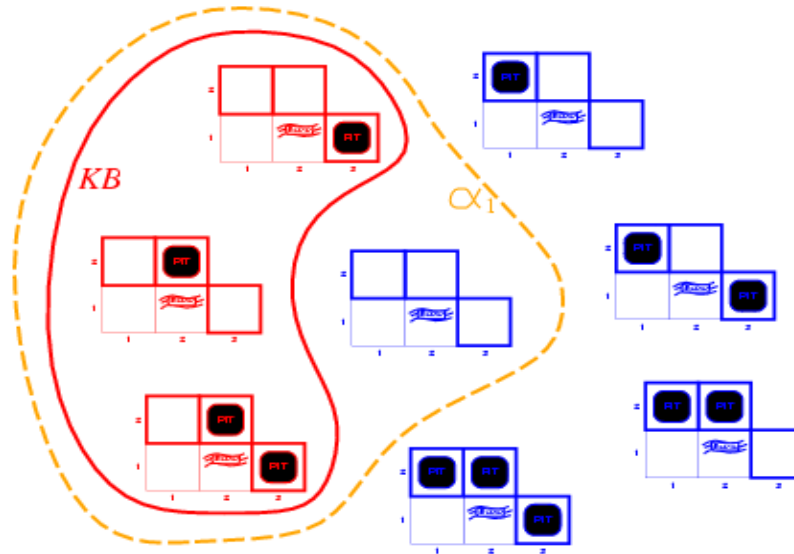
Wumpus models

$\alpha_1 = "[1,2] \text{ is safe}"?$



$KB = \text{wumpus-world rules} + \text{observations}$

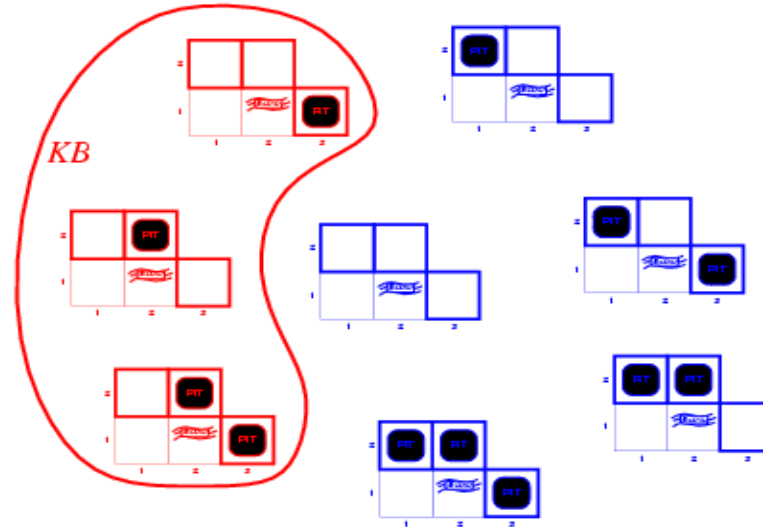
Wumpus models



$\alpha_1 = "[1,2] \text{ is safe}]", KB \models \alpha_1$, proved by model checking

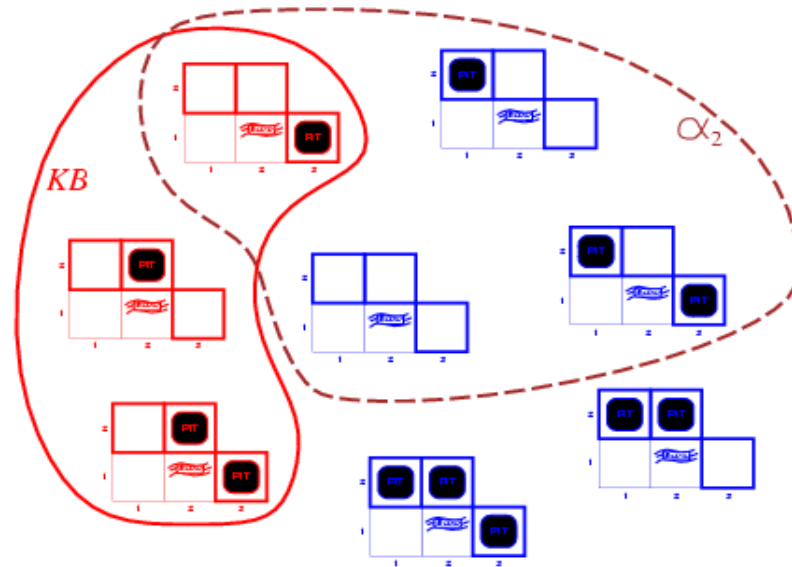
Wumpus models

$\alpha_2 = "[2,2]$ is safe“?



KB = wumpus-world rules + observations

Wumpus models



$\alpha_2 = "[2,2] \text{ is safe}]", KB \not\models \alpha_2$

Inference

- $KB \vdash_i \alpha$ = sentence α can be derived from KB by procedure i
- **Soundness**: i is sound if whenever $KB \vdash_i \alpha$, it is also true that $KB \models \alpha$
- **Completeness**: i is complete if whenever $KB \models \alpha$, it is also true that $KB \vdash_i \alpha$
- Preview: we will define a logic (first-order logic) which is expressive enough to say almost anything of interest, and for which there exists a sound and complete inference procedure.
- That is, the procedure will answer any question whose answer follows from what is known by the KB .

- **Soundness** (of an algorithm) means that the algorithm doesn't yield any results that are untrue. If, for instance, I have a sorting algorithm that sometimes does not return a sorted list, the algorithm is not sound.
- **Completeness** means that the algorithm addresses all possible inputs and doesn't miss any. So, if my sorting algorithm never returned an unsorted list, but simply refused to work on lists that contained the number 7, it would not be complete.

Propositional logic: Syntax

- Propositional logic is the simplest logic – illustrates basic ideas
- The proposition symbols P_1, P_2 etc are sentences
 - If S is a sentence, $\neg S$ is a sentence (**negation**)
 - If S_1 and S_2 are sentences, $S_1 \wedge S_2$ is a sentence (**conjunction**)
 - If S_1 and S_2 are sentences, $S_1 \vee S_2$ is a sentence (**disjunction**)
 - If S_1 and S_2 are sentences, $S_1 \Rightarrow S_2$ is a sentence (**implication**)
 - If S_1 and S_2 are sentences, $S_1 \Leftrightarrow S_2$ is a sentence (**biconditional**)

Propositional logic: Semantics

Each model specifies true/false for each proposition symbol

E.g. $P_{1,2}$ false $P_{2,2}$ true $P_{3,1}$ false

Rules for evaluating truth with respect to a model m :

$\neg S$	is true iff	S is false	
$S_1 \wedge S_2$	is true iff	S_1 is true and	S_2 is true
$S_1 \vee S_2$	is true iff	S_1 is true or	S_2 is true
$S_1 \Rightarrow S_2$	is true iff	S_1 is false or	S_2 is true
i.e.,	is false iff	S_1 is true and	S_2 is false
$S_1 \Leftrightarrow S_2$	is true iff	$S_1 \Rightarrow S_2$ is true and	$S_2 \Rightarrow S_1$ is true

Simple recursive process evaluates an arbitrary sentence, e.g.,

$$\neg P_{1,2} \wedge (P_{2,2} \vee P_{3,1}) = \text{true} \wedge (\text{true} \vee \text{false}) = \text{true} \wedge \text{true} = \text{true}$$

Truth tables for connectives

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
false	false	true	false	false	true	true
false	true	true	false	true	true	false
true	false	false	false	true	false	false
true	true	false	true	true	true	true

Wumpus world sentences

Let $P_{i,j}$ be true if there is a pit in $[i, j]$.

Let $B_{i,j}$ be true if there is a breeze in $[i, j]$.

$\neg P_{1,1}$

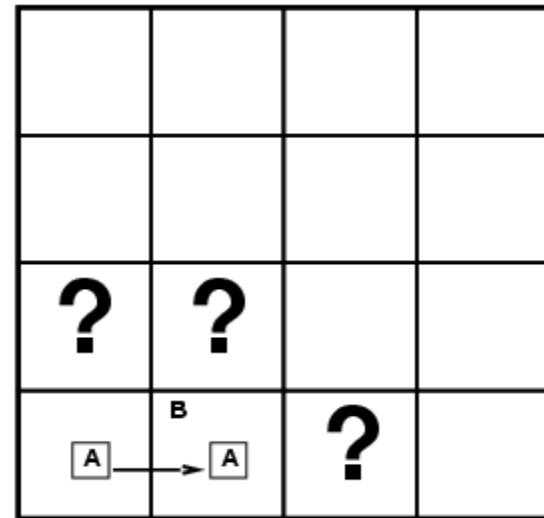
$\neg B_{1,1}$

$B_{2,1}$

- "Pits cause breezes in adjacent squares"

$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

$B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$



Logical equivalence

- Two sentences are **logically equivalent** iff true in same models: $\alpha \equiv \beta$
iff $\alpha \models \beta$ and $\beta \models \alpha$

$$(\alpha \wedge \beta) \equiv (\beta \wedge \alpha) \quad \text{commutativity of } \wedge$$

$$(\alpha \vee \beta) \equiv (\beta \vee \alpha) \quad \text{commutativity of } \vee$$

$$((\alpha \wedge \beta) \wedge \gamma) \equiv (\alpha \wedge (\beta \wedge \gamma)) \quad \text{associativity of } \wedge$$

$$((\alpha \vee \beta) \vee \gamma) \equiv (\alpha \vee (\beta \vee \gamma)) \quad \text{associativity of } \vee$$

$$\neg(\neg\alpha) \equiv \alpha \quad \text{double-negation elimination}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\beta \Rightarrow \neg\alpha) \quad \text{contraposition}$$

$$(\alpha \Rightarrow \beta) \equiv (\neg\alpha \vee \beta) \quad \text{implication elimination}$$

$$(\alpha \Leftrightarrow \beta) \equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) \quad \text{biconditional elimination}$$

$$\neg(\alpha \wedge \beta) \equiv (\neg\alpha \vee \neg\beta) \quad \text{de Morgan}$$

$$\neg(\alpha \vee \beta) \equiv (\neg\alpha \wedge \neg\beta) \quad \text{de Morgan}$$

$$(\alpha \wedge (\beta \vee \gamma)) \equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) \quad \text{distributivity of } \wedge \text{ over } \vee$$

$$(\alpha \vee (\beta \wedge \gamma)) \equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) \quad \text{distributivity of } \vee \text{ over } \wedge$$

7 Inference Rules for Propositional Logic

- Modus Ponens

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

- And-Elimination

$$\frac{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n}{\alpha_i}$$

- And-Introduction

$$\frac{\alpha_1, \alpha_2, \dots, \alpha_n}{\alpha_1 \wedge \alpha_2 \wedge \dots \wedge \alpha_n}$$

- Or-Introduction

$$\frac{\alpha_i}{\alpha_1 \vee \alpha_2 \vee \dots \vee \alpha_n}$$

- Double-Negation-Elimination

$$\frac{\neg(\neg\alpha)}{\alpha}$$

- Unit Resolution

$$\frac{\alpha \vee \beta, \neg\beta}{\alpha}$$

- Resolution

$$\frac{\alpha \vee \beta, \neg\beta \vee \gamma}{\alpha \vee \gamma}$$

$$\alpha \vee \gamma$$

atau

$$\frac{\neg\alpha \Rightarrow \beta, \beta \Rightarrow \gamma}{\neg\alpha \Rightarrow \gamma}$$

$$\neg\alpha \Rightarrow \gamma$$



PR: Menurunkan Sentence α dari KB menggunakan Logical Equivalence dan Inference



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Validity and satisfiability

A sentence is **valid** if it is true in **all** models,
e.g., *True*, $A \vee \neg A$, $A \Rightarrow A$, $(A \wedge (A \Rightarrow B)) \Rightarrow B$

Validity is connected to inference via the **Deduction Theorem**:
 $KB \models \alpha$ if and only if $(KB \Rightarrow \alpha)$ is valid

A sentence is **satisfiable** if it is true in **some** model
e.g., $A \vee B$, C

A sentence is **unsatisfiable** if it is true in **no** models
e.g., $A \wedge \neg A$

Satisfiability is connected to inference via the following:
 $KB \models \alpha$ if and only if $(KB \wedge \neg \alpha)$ is unsatisfiable

- Proof methods divide into (roughly) two kinds:
 - **Inference rules**
 - Legitimate (sound) generation of new sentences from old
 - **Proof** = a sequence of inference rule applications
Can use inference rules as operators in a standard search algorithm
 - Typically require transformation of sentences into a **normal form**
 - **Model checking**
 - truth table enumeration (always exponential in n)
 - improved backtracking, e.g., Davis--Putnam-Logemann-Loveland (DPLL)
 - heuristic search in model space (sound but incomplete)
e.g., min-conflicts-like hill-climbing algorithms

Expressiveness limitation of propositional logic

- KB contains "physics" sentences for every single square
- For every time t and every location $[x,y]$,
$$L_{x,y}^t \wedge FacingRight^t \wedge Forward^t \Rightarrow L_{x+1,y}^{t+1}$$
- Rapid proliferation of clauses

Summary

- Logical agents apply **inference** to a **knowledge base** to derive new information and make decisions
- Basic concepts of logic:
 - **syntax**: formal structure of **sentences**
 - **semantics**: **truth** of sentences wrt **models**
 - **entailment**: necessary truth of one sentence given another
 - **inference**: deriving sentences from other sentences
 - **soundness**: derivations produce only entailed sentences
 - **completeness**: derivations can produce all entailed sentences
- Wumpus world requires the ability to represent partial and negated information, reason by cases, etc.
- Propositional logic lacks expressive power